LEF MODEL · CHAPTER 17

Chapter 17: The Pacific Stripes — the Front Wall of the Pear and the Mirror Imprint of the Moon

Based on Paper 09 — The Stripes: The leading side of the pear on the floor of the Pacific (Parts A–C.4: the parallel latitude lines, the two knees in the west, the mirror imprint at 1.37× and the length of the pear)

01 · THE MYSTERY

The mystery: why do six straight, parallel stripes thousands of kilometres long cross the floor of the Pacific — and converge in the west?

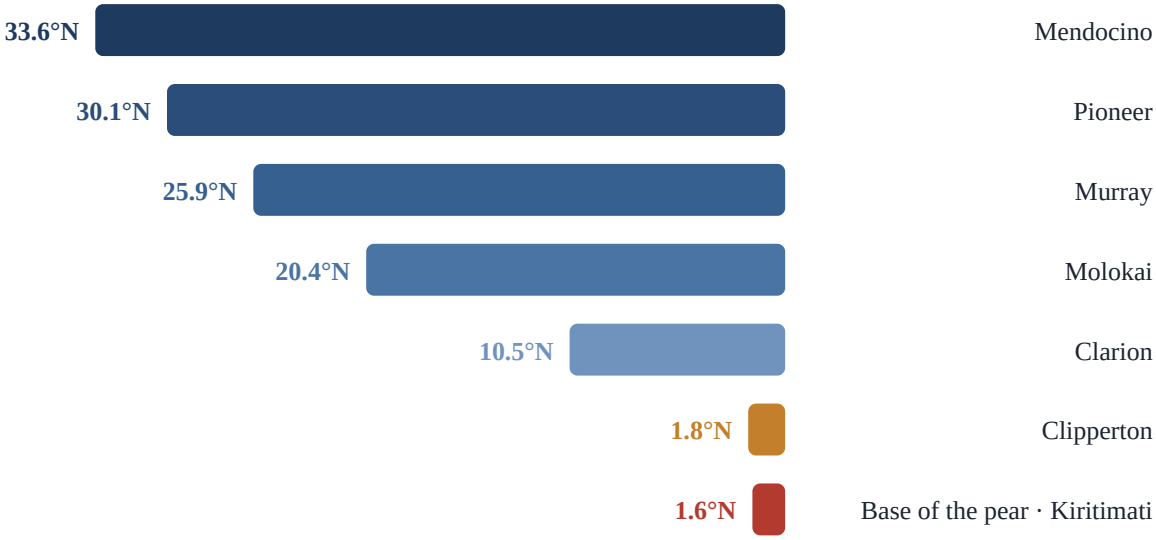
In the vertical gravity gradient maps (VGG — Vertical Gravity Gradient) of the North Pacific, an extraordinary geometric system stands out: **six straight, sharp and remarkably parallel stripes** — Mendocino, Pioneer, Murray, Molokai, Clarion and Clipperton — stretching west from the shores of North America for thousands of kilometres (Clipperton alone extends over about **7,240 km**, more than the Earth's radius!), and many of them appear double or triple. In this chapter we will discover four surprising geometric phenomena indicating that these stripes are **the scar of the pear's return**, the pear from which the Moon was born:

1. **Parallels of one shared equator (A.1, B.1):** the six stripes are not parallel to today's equator, but are parallel latitude lines of **one ancient equator** tilted by ≥ 10° relative to today's equator — spread from 33.6°N (Mendocino) to 1.8°N (Clipperton), plus two more southern stripes (6°S and 14°S).

2 **The base of the pear at Kiritimati sits exactly on the system's equator (B.2–B.3):** the island of Kiritimati (1.9°N, 157.5°W, identified in Paper 3 as the base of the pear above the lower mantle's large low-velocity province, LLVP) sits only **1.6°** from the stripes' equator — and the stripes run from it **eastward only** (toward America)!

3 **Two "knees" in the west bounding a width of 3,360 km (C.1–C.2):** west of Kiritimati the stripes leave their parallel course: the northern stripe bends down (23.8°N, 164.9°E) and the southern stripe bends up (5.3°S, 173.2°E). The distance between the two knees is **3,360 km** (range 3,040–3,700 km) — exactly the width of the Moon's contact zone (3,000 km) and the dome above it (3,650 km) — and their centre (9.3°N, 169.3°E) lies **3,760 km west of the base**!

4 **A mirror imprint of the lunar maria, enlarged 1.37 times, inside the tip (C.3):** within the footprint of the tip in the west appears an enlarged mirror image of the Moon's near side, in which Imbrium (13.19°N, 179.04°E), Serenitatis (11.5°N, 166.3°E) and the contact centre (8.2°N, 179.3°W) all fall within the radius of the tip (1,680 km)!



The latitude of each stripe relative to the equator of the stripe system: Clipperton and the base of the pear at Kiritimati sit almost on the same line.

02 · AN IMAGE

An image: a fold in a tablecloth under a book, a seal in wax, and a tree lying on the ground

Four everyday images illustrate how a giant pear that grew out of the Earth and was drawn back inward after the Moon's release left behind the stripes of the Pacific:

- **A child walking outward on a spinning merry-go-round (the pear tilting back toward the trailing side, A.2):** while the pear stretched out of the Earth rotating eastward, the material climbing upward needed a greater orbital speed and so *lagged behind* against the direction of rotation — that is, **westward**. So the pear got two walls: **the front wall (Leading Side, in the east)**, which stretched in the direction of rotation, and **the back wall (Trailing Side, in the west)**, beneath the tilted tip!
- **A fold in a tablecloth under a book (the formation of the stripes from below and from the sides, B.5):** when the Moon detached at 3.873 Ga, "the cork was pulled", the internal pressure in the pear was released, and the pear was drawn in and absorbed back into the Earth. The stretched front wall was smeared eastward in the direction of rotation and folded on itself like the folds of a tablecloth pushed from the sides: that is why the stripes **pass beneath the volcanic ridges without breaking them** (unlike a groove carved from above, which would have cut the ridge!), while the unstretched back wall collapsed into a heap and crumpled into the twisting island arcs of the western Pacific (the Philippines, the Marianas, Indonesia and the Banda arc, B.8)!
- **A rubber stamp pressed into soft wax (the mirror imprint of the Moon's near side, C.3):** face to face, the near side of the Moon sat on the tip of the pear. Like letters carved in reverse on a rubber stamp, or a palm pressed against a window pane from inside and seen from outside, north stays north, but **east and west swap** — and since the Moon's density on the tip was 1,300 kg/m³ against 3,344 kg/m³ today, every dimension in the imprint is $(3344/1300)^{1/3} = 1.37$ times larger!

- **A felled tree lying on the ground along its full length (the match of the 9,360 km length with the length of the pear, C.4):** the pear did not fall like a tree but was drawn inward, yet the front wall smeared over the ocean floor extended over a length identical to the length of the pear's body itself at the moment of detachment!

03 · SCIENCE AND MODEL

Science and model: the stripe system, the two knees and the mirror imprint (Parts A–C.4)

Parts A–C.4 of Paper 09 present four independent geometric measurements that confirm the identity of the stripes as the front wall of the pear and the knee area as the place where the Moon formed:

1. **The order of the stripes relative to the system's equator and to the base (A.1, B.1–B.4):** on a three-dimensional globe (Cesium) calibrated to known reference points (Hawaii, the Hawaii–Emperor bend, the tip of the Tonga trench and the Meiji seamount, to an accuracy of < 5 pixels), the six stripes keep a constant distance from one shared equator: Mendocino (33.6°N), Pioneer (30.1°N), Murray (25.9°N), Molokai (20.4°N), Clarion (10.5°N) and Clipperton (1.8°N — 0.2°–0.8° from the latitude of Kiritimati, which sits at 1.6°N!). South of the base two more stripes were identified (about 6°S and 14°S, probably Galápagos and Marquesas), proving that the base of the pear (9,000–11,000 km wide) stood *within* the width of the front wall and not at its edge.

2. **The continuity of the stripes beneath the ridges and their convergence in the west (B.5–B.6):** at four separate intersections (near 36°N, 178°W, at the Line Islands 6°N, 159°W, at the Phoenix Islands 2°S, 175°W and at 13°S, 152°W) the stripe goes beneath the volcanic ridge and continues on the other side with no sideways jump and without cutting the ridge. West of the ridges, the northern stripe bends south and the southern stripe bends north toward a shared area of convergence.

3 **The mirror imprint of the near side at × 1.37 magnification (C.3):** after the author replaced an initial direct reading (in which Nectaris landed 2,500 km from its place) with the physically correct mirror reading (swapping east–west and enlarging 1.37 times, so that the Imbrium–Serenitatis distance grows from 1,023 km on the Moon today to ~ 1,400 km), anchoring **Imbrium** at 13.19°N, 179.04°E places **Serenitatis** (11.5°N, 166.3°E) on the Marshall chain only 410 km from the centre of the tip, and **the contact centre** (8.2°N, 179.3°W) 1,260 km away — all deep within the radius of the tip (1,680 km), while Nectaris (1,970 km) falls just outside the edge, exactly like its southern position on the Moon!

04 · THE MECHANISM IN WORDS

The mechanism in words: how to compute the mirror scale (1.37×) and the length of the pear's body without the Moon

Parts C.1–C.4 of Paper 09 show how two simple geometric properties — conservation of the Moon's mass during its compaction, and the position of the final contact relative to the corotation line — determine in advance, with no free parameters, both the size of the mirror imprint on the ocean floor and the length of the whole stripe system:

1 **The mirror magnification factor (× 1.37, §C.3):** while the young Moon sat on the tip of the pear, it was swollen and rich in gases and volatiles, with a mean density of $\rho_{\text{tip}} = 1,300$ kg/m³. Since then it has compacted to its density today ($\rho_{\text{today}} = 3,344$ kg/m³, radius 1,737 km) while keeping its mass constant. Since the volume of a sphere is inversely proportional to its density and the radius is proportional to the cube root of the volume, every distance on the face of the young Moon was **1.37** times larger than its value today (and by the series' two-step calculation — a transition from lens to sphere by a factor of 0.83 and then shrinkage from 1,890 to 1,737 km — one gets 1.31, a match within a few percent). So the distance between Imbrium and Serenitatis (1,023 km today) was **≈ 1,400 km** on the tip of the pear!

2. **Separating the length of the pear's body from the length of the Moon itself (§C.4):** at the moment of detachment (an Earth day of 5.77 hours), the corotation line at the detachment latitude (22.5°) stood 17,287 km from the Earth's centre. The Moon's centre (its radius at detachment being 1,890 km) crossed the line by about 210 km and stood at 17,497 km. Therefore:
    - 3. **The pear together with the Moon as its tip:** reached exactly to the corotation line — a distance of 17,287 − (6,602 to 6,721) = 10,570–10,690 km from the Earth's surface.
    - 4. **The pear's body itself (including the tip area, without the Moon, which detached and left):** ended one lunar radius inward of the Moon's centre (17,497 − 1,890 = 15,607 km from the Earth's centre), that is, at a length of **8,890–9,000 km** from the Earth's surface — almost identical to the length of the stripe system spread over the ocean floor from the centre of the tip to Clipperton in the east (**9,360 km**)!
5. **The alignment of the knees on the system's lines (§C.1–C.2):** the two knees are not random: the northern knee (23.8°N, 164.9°E) sits exactly on the latitude of the Pioneer stripe (30.0°N from the system's equator, against 30.1°N), and the southern knee (5.3°S, 173.2°E) sits exactly on the system's equator itself (0.1° from it) — the same equator on which the base of the pear at Kiritimati sits (1.6° from it)!

05 · THE KEY EQUATIONS

Academic explanation and the key equations

The ratio of distances on the Moon's face between the tip stage and the present, the length of the pear's body remaining on the Earth, and the radius of the tip's footprint are formulated as follows (§C.1, §C.3–C.4):

$$\frac{L_{\text{tip}}}{L_{\text{today}}} = \left(\frac{\rho_{\text{today}}}{\rho_{\text{tip}}}\right)^{1/3} = \left(\frac{3344}{1300}\right)^{1/3} = 1.37$$

$$L_{\text{pear body}} = (R_{\text{corot},\,22.5^\circ} + \Delta r_{\text{release}} - R_{\text{Moon,\,rel}}) - R_{\oplus} = 15{,}607 - (6{,}602$$

Where $\Delta r_{\text{release}} \approx 210$ km is the distance beyond the corotation line (17,287 km at latitude 22.5°) and $R_{\text{Moon,\,rel}} = 1,890$ km is the Moon's radius at the moment of detachment, it follows that the length of the pear's body (8,890–9,000 km) matches to within 4%–5% the length of the stripe system (9,360 km) and the arcs of the back side (8,100–10,300 km).

06 · THE SIGNATURE

The signature in reality: the map of latitude lines and the mirror coordinates on the Pacific floor

Besides the sharp gravity-gradient (VGG) lines crossing the North Pacific with no sideways deviation over 30° of latitude, four precise geographic anchors hold on the map (§A.1, §B.7, §C.3):

- **The deep anchor in the lower mantle (B.7):** beneath the base of the pear at Kiritimati (1.9°N, 157.5°W), the lowest layer of the mantle is seismically slow in all five vote models of Cottaar & Lekic (2016) and lies within the −1% contour of the Pacific large low-velocity province (LLVP — its centre 1,480–1,660 km south-south-west of the base), into which the heavy, iron-rich material of the pear sank as it was drawn back inward.
- **The mirror imprint within the tip's footprint (radius ≈ 1,680 km around 9.3°N, 169.3°E):** three of the four key points of the Moon's near side fall deep within the mirror footprint — Serenitatis (410 km from the centre), Imbrium (1,150 km) and the contact centre (1,260 km), while Nectaris (1,970 km) lies just outside the southern edge, exactly like its position relative to the contact zone on the Moon!

The lab box: the latitudes of the stripes, the mirror imprint (× 1.37) and the lengths of the pear (Paper 09, Parts A–C.4)

Stripe / geographic anchor	Distance from the system's equator	Distance north of the base latitude (1.6°N)	Geometric role in the structure of the pear
1. Mendocino	**33.6°N**	**32.0° north**	**The northern boundary of the stripe system**
2. Pioneer	30.1°N	28.5° north	Matches the latitude of the northern knee (30.0°N)
3. Murray	25.9°N	24.3° north	A central northern stripe
4. Molokai	20.4°N	18.8° north	A central stripe
5. Clarion	10.5°N	8.9° north	A south-central stripe
6. Clipperton	**1.8°N** (0.8° at the western end)	**0.2° north** (length ~ 7,240 km)	**Passes right alongside the base of the pear**
The base of the pear (Kiritimati)	**1.6°N** (1.9°N, 157.5°W)	**0.0° (the reference point)**	**Sits on the stripes' equator and above the LLVP (width 9,000–11,000 km)**
Two southern stripes (Galápagos / Marquesas)	~ 6°S and ~ 14°S	7.6° and 15.6° south	Prove that the base stood within the width of the front wall

Object / measurement (Part C)	Coordinates on the Moon	Predicted coordinates on the Earth (× 1.37, mirror)	Distance from the centre of the tip (9.3°N, 169.3°E)	Fit to the footprint (R ≈ 1,680 km)
Imbrium (the position anchor)	**37.6°N, 340.0°E**	**13.19°N, 179.04°E**	**1,150 km (2,870 km from the base)**	**Inside the footprint (the main passage)**
Serenitatis (predicted in the mirror)	**27.3°N, 18.4°E**	**≈ 11.5°N, 166.3°E**	**410 km (4,140 km from the base)**	**Deep inside the footprint (on the Marshall chain)**
The contact centre (predicted in the mirror)	**24.1°N, 335.2°E**	**≈ 8.2°N, 179.3°W**	**1,260 km (2,510 km from the base)**	**Inside the footprint**

Nectaris (predicted in the mirror)	15.2°S, 35.0°E	≈ 3.1°S, 156.6°E	1,970 km (5,130 km from the base)	Outside the southern edge (matches its southern position on the Moon)

07 · NEXT

The door ahead

In this chapter we saw how the six parallel stripes of the North Pacific line up on one ancient equator, run east from the base of the pear at Kiritimati, and converge in the west between two knees 3,360 km apart, within which the enlarged (× 1.37) mirror image of the Moon's near side is imprinted.

Chapter 18

The Convergence of the Whole Pacific, the Spatula Ridges and the Solution to the Riddle of the Floor's Young Age →

← Chapter 16 · Prologue

© 2026 Zohar Duran · The LEF Model · lef-model.com · ORCID 0009-0005-3194-9387